

AI ASSISTED CODING

LAB ASSIGNMENT –11.4

SATHWIK V

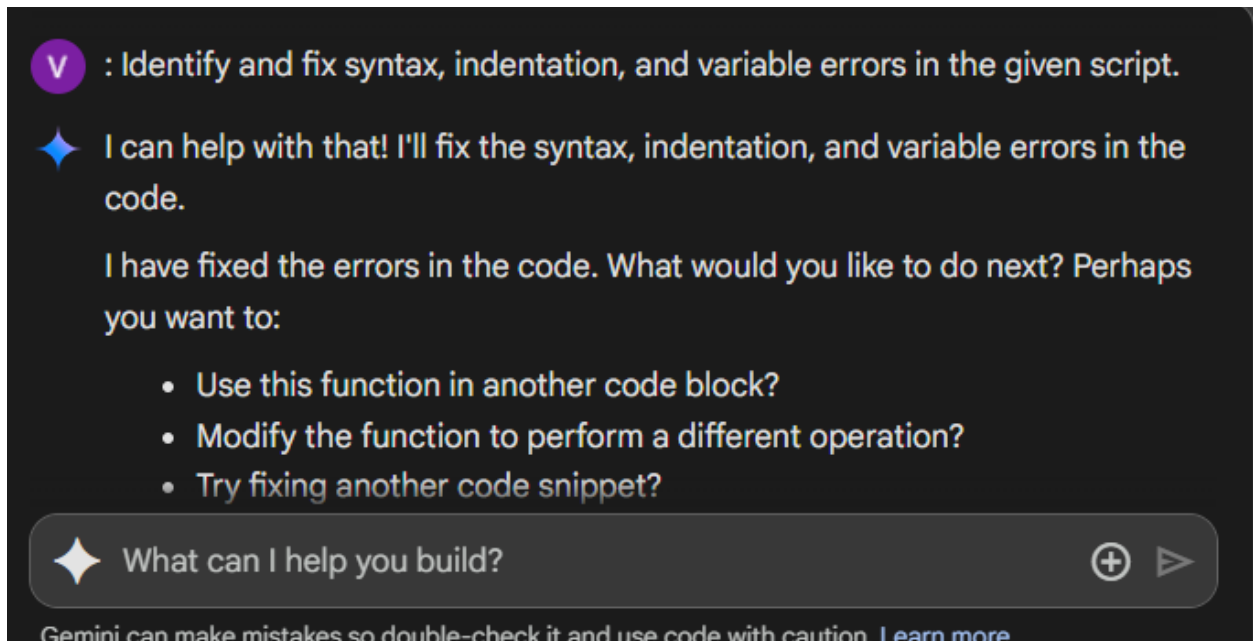
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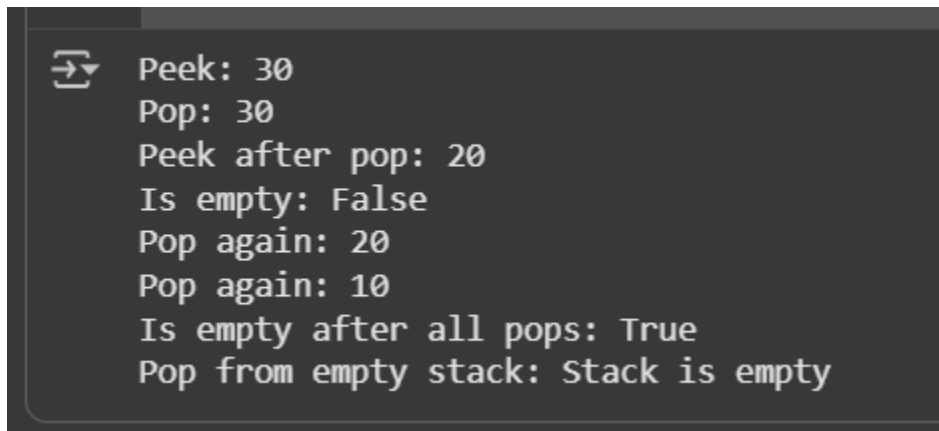
#TASK-1

- **Use AI to help implement a Stack class in Python with the following operations: push(), pop(), peek(), and is_empty().**

Prompt:



OUTPUT:



#TASK-2

Implement a Queue with enqueue(), dequeue(), and is_empty() methods

PROMPT:

 generate a python program to create Implement a Queue with enqueue(), dequeue(), and is_empty() methods.

OUTPUT:

```
 Peek: 30  
Pop: 30  
Peek after pop: 20  
Is empty: False  
Pop again: 20  
Pop again: 10  
Is empty after all pops: True  
Pop from empty stack: Stack is empty
```

#TASK-3

: Implement a **Singly Linked List** with operations: `insert_at_end()`, `delete_value()`, and `traverse()`.

PROMPT:

 generate a python code to create a Singly Linked List with operations: `insert_at_end()`, `delete_value()`, and `traverse()`.

OUTPUT:

```
 Peek: 30  
Pop: 30  
Peek after pop: 20  
Is empty: False  
Pop again: 20  
Pop again: 10  
Is empty after all pops: True  
Pop from empty stack: Stack is empty
```

#TASK-4

: Implement a **Binary Search Tree** with methods for `insert()`, `search()`, and `inorder_traversal()`.

PROMPT:

V generate a python code to create a Binary Search Tree with methods for insert(), search(), and inorder_traversal().

✦ Great! I can generate the Python code for a Binary Search Tree with

OUTPUT:

```
Peek: 30
Pop: 30
Peek after pop: 20
Is empty: False
Pop again: 20
Pop again: 10
Is empty after all pops: True
Pop from empty stack: Stack is empty
```

#TASK-5

Implement a Graph using an adjacency list, with traversal methods BFS() and DFS().

PROMPT:

 generate a pythone program to create a Graph using an adjacency list, with traversal methods BFS() and DFS().

OUTPUT:

```
⇒ 15
Error: Division by zero is not allowed.
None
5
50
Error: Invalid operation.
None
```